Project 3

Exploring Keyword Co-occurrence Networks in Scientific Publications

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1. Project Objective

This project constructs a keyword co-occurrence network from a corpus of 66 academic articles, each annotated with up to 12 subject keywords. The objective is to uncover relationships between research themes by analyzing how frequently keywords appear together across articles.

2. Methodology

The pipeline begins with data preprocessing, where keyword columns are extracted, cleaned, and standardized to lowercase to ensure consistency.

Co-occurrence pairs are then generated for each article, treating the relationship as bidirectional so that both orderings of a pair contribute to the final edge weight. Pair frequencies are counted using a frequency distribution, producing a weighted adjacency matrix where each cell represents the number of times two keywords co-occur.

The adjacency matrix is subsequently loaded into a weighted network using NetworkX, where keywords serve as nodes and co-occurrence counts serve as edge weights. Node-level metrics — degree and strength — are computed to identify the most prominent and well-connected keywords in the corpus. Degree measures the number of distinct keywords a node is connected to, while strength captures the total co-occurrence intensity across all its connections. The network is visualized as a scatter plot of degree against strength, and the top co-occurring keyword pairs are ranked and reported.

3. Results

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Network summary:
  Nodes (keywords) : 248
  Edges (pairs)    : 2141
  Density          : 0.0699

Top 15 keywords by weighted degree:
                                  Weighted_Degree  Betweenness  PageRank
organizational behavior                       265     0.208043  0.046239
organizational effectiveness                  144     0.105075  0.025300
management science                            136     0.086903  0.024188
personnel management                          126     0.072894  0.022256
decision making                               112     0.092395  0.020645
organizational structure                      107     0.039925  0.018762
organizational sociology                       96     0.026746  0.016543
corporate governance                           85     0.024082  0.014946
industrial management                          84     0.027568  0.014835
strategic planning                             80     0.047741  0.014654
teams in the workplace                         78     0.017960  0.013537
organizational change                          60     0.015325  0.010451
industrial relations                           59     0.022675  0.010854
executive ability (management)                 57     0.015353  0.010037
management                                     53     0.009334  0.009132
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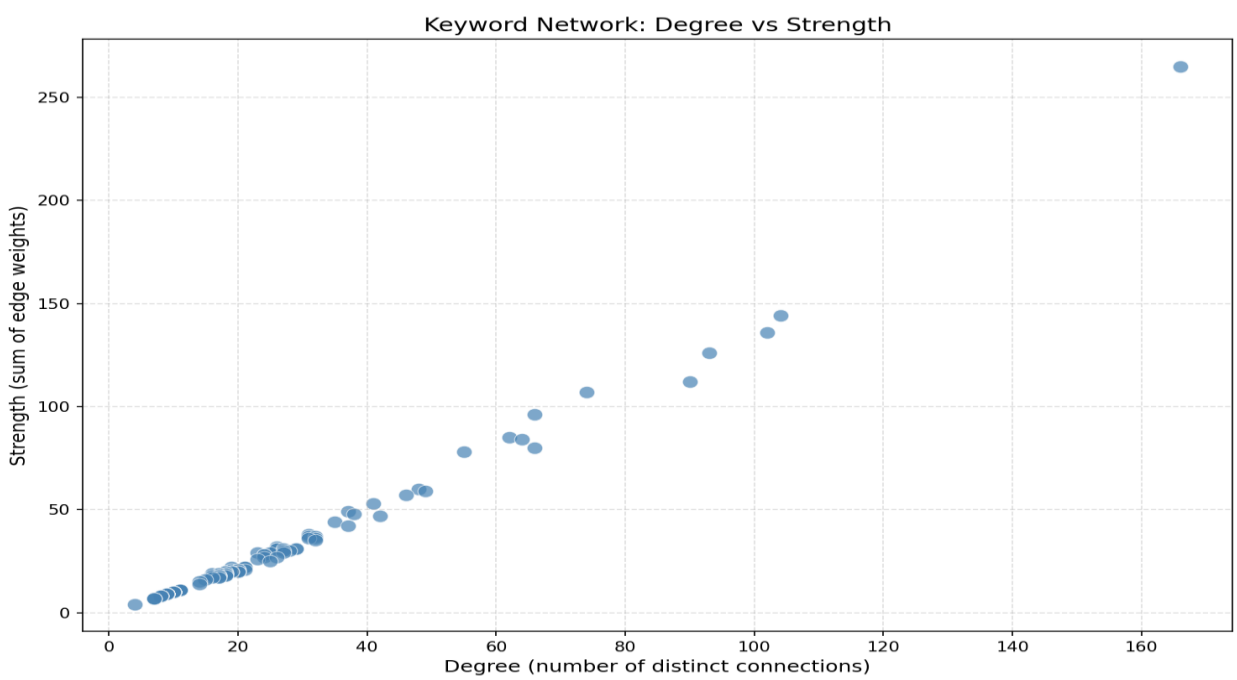
	Top 10 by Degree	Degree	Top 10 by Strength	Strength
1	organizational behavior	166	organizational behavior	265
2	organizational effectiveness	104	organizational effectiveness	144
3	management science	102	management science	136
4	personnel management	93	personnel management	126
5	decision making	90	decision making	112
6	organizational structure	74	organizational structure	107
7	organizational sociology	66	organizational sociology	96
8	strategic planning	66	corporate governance	85
9	industrial management	64	industrial management	84
10	corporate governance	62	strategic planning	80

	Keyword A	Keyword B	Weight
1	organizational behavior	organizational effectiveness	11
2	organizational behavior	organizational structure	9
3	organizational behavior	personnel management	8
4	management science	organizational behavior	7
5	organizational effectiveness	organizational structure	6
6	organizational behavior	organizational sociology	6
7	corporate governance	organizational behavior	6
8	decision making	organizational behavior	6
9	organizational behavior	organizational change	5
10	industrial relations	organizational behavior	5



"Organizational Behavior" emerges as the most central node, connecting to over 65% of all keywords (Degree = 166) and recording the highest co-occurrence intensity (Strength = 265). Its strongest linkage is with **"Organizational Effectiveness"**, the most frequent keyword pair in the corpus (weight = 11), indicating these two themes have very strong co-occurrence in the literature. Notably, Degree and Strength exhibit a near-perfect

linear correlation, suggesting that broadly connected keywords are also the most frequently co-occurring reflecting a consistent and concentrated thematic structure across the 66 articles.

4. Conclusion

This analysis provides a structured view of the intellectual landscape of the corpus, highlighting the most central research themes and the strongest conceptual linkages between them.